COURSE NAME: MCA
Semester-2
Subject Name: Data Structures

DAY 1

.	**Detailed Schedule**
1.	Data types – primitive and non-primitive, abstract data type, concept of data structures, what is an algorithm?, introduction to linear and nonlinear data structures, introduction to calculation of time complexity (best case, worst case and average case) and spacecomplexity.
Essential Assignment	
1.	Write a program that receives an integer and then prints the sum of digits.
2.	Write a program to find the total of each rows and columns in a given matrix.
Desirable Assignment	
1.	Write a program to find the second highest element from a 2D array.

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 2

.	**Detailed Schedule**
1.	**Array:** Operations (row major and column major address calculations), basic concept, difference between linked list and array, storage representation
Essential Assignment	
1.	Write a menu driven program to perform following operations for an array a) Print b) Insert at a specific position c) Delete at a specific position d) Exit
Desirable Assignment	
1.	Program to reverse an array and store it in another array.

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 3

.	**Detailed Schedule**
1.	**stack:** Basic concept, storage representation (array and linked list), basic operations (push, pop, peep and change)
Essential Assignment	
1.	Write a menu driven program to perform following operations in stack using static data structure. Also display the updated value of Top in every operation. a) Push b) Pop c) Peep d) Count e) Display f) Exit
Desirable Assignment	
1.	Write a program to reverse a given number using stack.

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 4

.	**Detailed Schedule**
1.	Applications of stacks (polish and reverse polish expressions), arithmetic expression evaluation using stack.
Essential Assignment	
1.	Write a program to convert infix expression to postfix
Desirable Assignment	
1.	Write an algorithm to add and multiply of one variable polynomial

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 5

.	**Detailed Schedule**
1.	**Queues**: Basic concept, storage representation (array and linked list), basic operations (insert and delete).
Essential Assignment	
1.	Write a program to evaluate arithmetic expression from postfix expression.
Desirable Assignment	
1.	Write a program to implement FCFS algorithm for process using queue.

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 6

.	**Detailed Schedule**
1.	Types of queues- circular, deque, priority queues (only concept). application of queues.
Essential Assignment	
1.	Write a menu driven program to perform following operations in simple queue using static data structure. Also display the updated value of Front and Rear in every operation. a) Insert b) Delete c) Display d) Exit
Desirable Assignment	
1.	Write a program to convert infix expression to prefix

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 7

.	**Detailed Schedule**
1.	**Linked List: B**asic concept, difference between linked list and array, storage representation, types of linked lists (singly, doubly, circular),
Essential Assignment	
1.	Write a menu driven program to perform following operations in circular queue using static data structure. Also display the updated value of Front and Rear in every operation. a) Insert b) Delete c) Display d) Exit
Desirable Assignment	
1.	Write a program to count the number of occurrences of a user specified element in the singly linked list. If that element does not present give an appropriate message.

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 8

.	**Detailed Schedule**
1.	Operations on linked list (insert, modify, delete)
Essential Assignment	
1.	Write a program using singly linked list to perform following operations: a) Create b) Display c) Insert - First, Last, At a specified location d) Delete -First, Last, At a specified location e)
Desirable Assignment	
1.	Write a program to display all the nodes of a singly linked list which is divisible by 7. If not found, then display appropriate message.

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures

DAY 9

.	**Detailed Schedule**
1.	Operations on linked list (union, intersection, merge, sort, searching, reverse (by changing address not data)),
Essential Assignment	
1.	Perform the following operations on singly linked list using user-defined functions: 1. Count total nodes 2. Create a copy of the linked list 3. Merge
Desirable Assignment	
1.	Write a program to display all the even node of a singly linked list. If not found, then display appropriate message

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 10

.	**Detailed Schedule**
1.	Applications of linked list (polynomial operations – addition and multiplication),
Essential Assignment	
1.	Perform the following operations on singly linked list using user-defined functions: 1. Reverse a listed list 2. Search for a specific node 3. Sort
Desirable Assignment	
1.	Write a program to search an existing element in a singly linked list. If not found, then display appropriate message

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 11

.	**Detailed Schedule**
1.	Sparse matrices implementation using linked list **Trees** – Basic concept, terms associated with trees (Node, parent, child, link, root, leaf, level, height, indegree, outdegree, siblings)
Essential Assignment	
1.	Write a program using doubly linked list to perform following operations a) Create b) Display c) Insert (At all locations) d) Delete (At all locations) e) Count f) Exit
Desirable Assignment	
1.	Write a program using doubly linked list to perform following operations a) Copy b) Search

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 12

.	**Detailed Schedule**
1.	Storage representation – Linear and Linked, Conversion of General tree to Binary tree, Complete Binary tree, full binary tree
Essential Assignment	
1.	Implement Stack using linked list
Desirable Assignment	
1.	Write a program using doubly linked list to perform following operations a) Merge b) Intersection of two linked lists

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 13

.	**Detailed Schedule**
1.	BST operations, Tree traversals – In-order, Pre-order, Post-order, Types of Binary tree : BST (Practical – recursive), Expression tree
Essential Assignment	
1.	Implement simple queue using linked list
Desirable Assignment	
1.	Write a program to implement circular queue using linked list.

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 14

.	**Detailed Schedule**
1.	AVL Tree (without practical),2-3 tree, B tree(without practical)), **Graphs**-Basic concepts, technical terms associated with Graphs – Digraph, Weighted graph, adjacent vertices, self loop, parallel edges, simple graph, complete graph, isolated vertex
Essential Assignment	
1.	1) Write program to create and display binary tree. 2) Write a program for all the operations of binary search tree.
Desirable Assignment	
1.	Write a program for all the operations of binary search tree.

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 15

.	**Detailed Schedule**
1.	Degree of a vertex, connected graph-, Storage representation (Set representation, Adjacency matrix, Adjacency list), Graph Traversing algorithms- DFS and BFS. Spanning Tree
Essential Assignment	
1.	Write a program to create a graph using adjacency matrix and traverse it using DFS and BFS algorithm.
Desirable Assignment	
1.	Write the program to do all operation of 2-3 tree.

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 16

.	**Detailed Schedule**
1.	Minimum spanning tree algorithm: Prim's Algorithm, Kruskal's Algorithm, Shortest path algorithm – Dijikstra 's algorithm
Essential Assignment	
1.	1) Write a program to find minimum spanning tree using prim's algorithm. 2) Write a program to find minimum spanning tree using Kruskal's algorithm.
Desirable Assignment	
1.	Find the BFS & DFS sequence for the graph by entering the graph using Linked list.

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 17

.	**Detailed Schedule**
1.	**Sorting Algorithms:** Bubble, Selection, Insertion, Shell, 2- Way Merge sort, Radix sort, heap sort and Quick sort.
Essential Assignment	
1.	1) Write an algorithm for following sorting techniques. a) Bubble b) Selection c) Insertion d) Shell 2) Consider the following data perform sorting such as bubble, selection, insertion, shell. 1,4,6,2,3,8,7,10,9
Desirable Assignment	
1.	Consider the following data perform sorting such as bubble, selection, insertion, shell. **mon, tue, wed, thu, fri, sat, sun**

COURSE NAME: MCA
Semester-2
Subject Name: Data Structures
DAY 18

.	**Detailed Schedule**
1.	Binary Search, **Hashing:** Introduction to Hashing and hashing methods, collision resolution techniques.
Essential Assignment	
1.	Program for Binary search, closed hashing, separate hashing.
Desirable Assignment	
1.	Consider the following data perform sorting such as 2-way merge sort, radix sort, heap sort, quick sort. **mon, tue, wed, thu, fri, sat, sun** **20,3,21,32,53,23,65,34,23**